SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1513

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**

Question Count: 5

Duration : 60 Minutes

Code of Conduct

I __J.P.BADRI(192524301)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action. Signature

of the student: ___ J.P.B _____

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Equivalent Steps in VMware Workstation Pro

- Library pane showing multiple VMs = multiple instances/regions.
- VM → Snapshot → Take Snapshot, then Snapshot Manager = backup/versioning demonstration.

- Edit → Virtual Network Editor to show NAT/Host-only/Bridged networks = hybrid or multi-cloud connectivity.
- VM → Manage → Clone = replica/auto-scaling demonstration.

Tip: Paste each screenshot directly under the relevant question in this document (Insert → Picture in Word) before final submission, with a one-line caption explaining what it demonstrates.

Analytical Problem Solving Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 – Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1. Compare Type-1 and Type-2 Hypervisors

Feature	Type-1 Hypervisor	Type-2 Hypervisor
Execution	Runs directly on hardware	Runs on a host operating system
Performance	High performance	Lower performance due to overhead
Security	Smaller attack surface	Depends on host OS security
Manageability	Centralized enterprise management	Simple desktop management
Examples	VMware ESXi, Microsoft Hyper-V	VMware Workstation Pro, Oracle VirtualBox

Analysis: Type-1 hypervisors provide better resource utilization, isolation, and reliability for mission-critical applications. Type-2 hypervisors introduce additional overhead because they depend on a host operating system.

Conclusion: Type-1 hypervisors are the better choice for cloud datacenters because they provide high performance, stronger isolation, and centralized management.

2. Workload Isolation in a Cloud Environment

Virtualization allows finance, healthcare, and student portal workloads to run on the same physical server using separate virtual machines.

Isolation mechanisms:

1. Each VM has separate virtual CPU, memory, and storage resources.
2. Hypervisor memory protection prevents unauthorized access between VMs.
3. Virtual networks and firewalls separate communication between workloads.
4. Role-based access control restricts administrative privileges.

Two major attack surfaces:

- Hypervisor vulnerabilities: An attacker may exploit a flaw to access other VMs.
- Virtual network attacks: Improper network configuration may expose sensitive data or allow unauthorized communication.

Conclusion: Strong hypervisor security, network segmentation, encryption, and regular patching protect workloads and maintain data confidentiality.

3. Analysis of VM Host Utilization

Given data:

Metric	Value	Analysis
CPU utilization	92%	Very high CPU demand
Memory utilization	88%	Limited memory headroom
Storage I/O latency	35 ms	Possible storage bottleneck
Average VM boot time	170 seconds	Slow VM startup

Likely bottlenecks:

1. High CPU utilization causes scheduling delays and reduced VM responsiveness.
2. High memory usage may cause memory contention or swapping.
3. Storage latency delays disk reads and writes, increasing application response time.
4. Slow VM boot time may result from resource contention, storage delays, or excessive startup services.

Corrective actions:

- Add CPU resources or migrate workloads to less-utilized hosts.
- Increase RAM and optimize memory allocation.
- Upgrade storage to faster SSD/NVMe or improve storage IOPS.
- Use VM templates and reduce unnecessary startup services.
- Monitor performance and use load balancing to distribute workloads.

Conclusion: CPU, memory, and storage optimization can reduce contention, improve VM startup time, and increase overall datacenter performance.

4. Software Defined Datacenter (SDDC) vs Traditional Datacenter

An SDDC uses software to manage and automate compute, storage, and networking resources instead of relying mainly on manually configured hardware.

Component	Traditional Datacenter	Software Defined Datacenter
Compute	Physical server allocation	Compute virtualization and automated VM provisioning
Storage	Dedicated storage hardware	Software-defined storage pools
Network	Manual switch configuration	Software-defined networking and virtual switches
Management	Mostly manual	Centralized, policy-based automation
Scalability	Slower hardware-dependent expansion	Faster resource allocation and scaling

Analysis:

1. Compute virtualization enables multiple VMs to share physical servers efficiently.
2. Storage virtualization combines storage devices into flexible pools for easy allocation.
3. Network virtualization creates isolated virtual networks and enables rapid configuration.
4. Automation reduces manual work and speeds up application deployment.

Conclusion: SDDC improves agility, scalability, resource utilization, and operational efficiency through centralized software-based management.

5. Multi-Cloud Federation: Identity and Privacy Design

Problem: Different cloud providers may use incompatible identity formats, authentication methods, and access-control policies. These mismatches can cause login failures or unauthorized access.

Proposed secure design:

1. Federated identity: Use a trusted identity provider (IdP) to authenticate users across cloud providers.
2. Standard protocols: Use OpenID Connect (OIDC) or SAML 2.0 for secure identity federation.
3. Identity mapping: Map user attributes and roles consistently across providers.
4. Least privilege: Apply role-based access control and grant only the permissions required.
5. Privacy protection: Minimize shared personal information, encrypt data in transit, and protect identity tokens.
6. Monitoring: Enable centralized audit logging, token expiry, and regular access reviews.

Architecture:

